

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Let $g(x)$ be an antiderivative for $f(x)$. Then $\ln(1+(g(x))^2)$ is an antiderivative for

(A) $\frac{2f(x)g(x)}{1+(f(x))^2}$ (B) $\frac{2f(x)g(x)}{1+(g(x))^2}$ (C) $\frac{2f(x)}{1+(f(x))^2}$ (D) none

MII001

2. Consider $f(x) = \frac{x^2}{1+x^3}$; $g(t) = \int f(t)dt$. If $g(1) = 0$ then $g(x)$ equals -

(A) $\frac{1}{3}\ln(1+x^3)$ (B) $\frac{1}{3}\ln\left(\frac{1+x^3}{2}\right)$ (C) $\frac{1}{2}\ln\left(\frac{1+x^3}{3}\right)$ (D) $\frac{1}{3}\ln\left(\frac{1+x^3}{3}\right)$

MII002

3. $\int \frac{x^2(1-\ln x)}{\ln^4 x - x^4} dx$ equals

(A) $\frac{1}{2}\ln\left|\frac{x}{\ln x}\right| - \frac{1}{4}\ln|\ln^2 x - x^2| + C$ (B) $\frac{1}{4}\ln\left|\frac{\ln x - x}{\ln x + x}\right| - \frac{1}{2}\tan^{-1}\left(\frac{\ln x}{x}\right) + C$
 (C) $\frac{1}{4}\ln\left|\frac{\ln x + x}{\ln x - x}\right| + \frac{1}{2}\tan^{-1}\left(\frac{\ln x}{x}\right) + C$ (D) $\frac{1}{4}\left(\ln\left|\frac{\ln x - x}{\ln x + x}\right| + \tan^{-1}\left(\frac{\ln x}{x}\right)\right) + C$

MII003

4. Primitive of $\frac{3x^4 - 1}{(x^4 + x + 1)^2}$ w.r.t. x is -

(A) $\frac{x}{x^4 + x + 1} + c$ (B) $-\frac{x}{x^4 + x + 1} + c$ (C) $\frac{x+1}{x^4 + x + 1} + c$ (D) $-\frac{x+1}{x^4 + x + 1} + c$

MII004

5. $\int \frac{x^2 + x}{(e^x + x + 1)^2} dx$ equals

(A) $C - \ln(1 + (x+1)e^x) - \frac{1}{1 + (x+1)e^x}$ (B) $C - \ln(1 + (x+1)e^x) + \frac{1}{1 + (x-1)e^x}$
 (C) $C - \ln(1 - (x^2 - 1)e^{-x}) + \frac{1}{1 + (x+1)e^x}$ (D) $C - \ln(1 + (x+1)e^{-x}) - \frac{1}{1 + (x+1)e^{-x}}$

MII005

6. $\int \frac{1-x^7}{x(1+x^7)} dx$ equals -

(A) $\ln x + \frac{2}{7}\ln(1+x^7) + c$ (B) $\ln x - \frac{2}{7}\ln(1-x^7) + c$
 (C) $\ln x - \frac{2}{7}\ln(1+x^7) + c$ (D) $\ln x + \frac{2}{7}\ln(1-x^7) + c$

MII006

7. Integral of $\sqrt{1+2\cot x(\cot x+\operatorname{cosec} x)}$ w.r.t. x is

- (A) $2\ell n \cos \frac{x}{2} + c$ (B) $2\ell n \sin \frac{x}{2} + c$
 (C) $\frac{1}{2} \ell n \cos \frac{x}{2} + c$ (D) $\ell n \sin x - \ell n(\operatorname{cosec} x - \cot x) + c$

MII007

8. Let $f(x) = \frac{2\sin^2 x - 1}{\cos x} + \frac{\cos x(2\sin x + 1)}{1 + \sin x}$ then $\int e^x (f(x) + f'(x)) dx$

(where c is the constant of integration)

- (A) $e^x \tan x + c$ (B) $e^x \cot x + c$ (C) $e^x \operatorname{cosec}^2 x + c$ (D) $e^x \sec^2 x + c$

MII008

9. $\int e^x \left(\frac{x^2 - 3}{(x-1)^2} \right) dx$ is equal to $-$ (where C is constant of integration)

- (A) $e^x \frac{(x+3)}{(x-1)} + C$ (B) $e^x \left(\frac{x-3}{x-1} \right) + C$ (C) $e^x \left(\frac{x+1}{x-1} \right) + C$ (D) $e^x \left(\frac{1}{x-1} \right)^2 + C$

MII009

10. $\int x \cdot \frac{\ln(x + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx$ equals -

- (A) $\sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) - x + c$ (B) $\frac{x}{2} \cdot (x + \sqrt{1+x^2}) \ln^2 - \frac{x}{\sqrt{1+x^2}} + c$
 (C) $\frac{x}{2} \cdot \ln^2(x + \sqrt{1+x^2}) + \frac{x}{\sqrt{1+x^2}} + c$ (D) $\sqrt{1+x^2} \ln(x + \sqrt{1+x^2}) + x + c$

MII010

11. The evaluation of $\int \frac{p x^{p+2q-1} - q x^{q-1}}{x^{2p+2q} + 2x^{p+q} + 1} dx$ is

- (A) $-\frac{x^p}{x^{p+q} + 1} + C$ (B) $\frac{x^q}{x^{p+q} + 1} + C$ (C) $-\frac{x^q}{x^{p+q} + 1} + C$ (D) $\frac{x^p}{x^{p+q} + 1} + C$

MII011

12. The integral $\int \sqrt{\cot x} e^{\sqrt{\sin x}} \sqrt{\cos x} dx$ equals

- (A) $\frac{\sqrt{\tan x} e^{\sqrt{\sin x}}}{\sqrt{\cos x}} + C$ (B) $2e^{\sqrt{\sin x}} + C$ (C) $-\frac{1}{2} e^{\sqrt{\sin x}} + C$ (D) $\frac{\sqrt{\cot x} e^{\sqrt{\sin x}}}{2\sqrt{\cos x}} + C$

MII012

13. $\int (\sin(101x) \cdot \sin^{99} x) dx$ equals

- (A) $\frac{\sin(100x)(\sin x)^{100}}{100} + C$ (B) $\frac{\cos(100x)(\sin x)^{100}}{100} + C$
 (C) $\frac{\cos(100x)(\cos x)^{100}}{100} + C$ (D) $\frac{\sin(100x)(\sin x)^{101}}{101} + C$

MII013

Indefinite Integration

14. $\int \sec^2 \theta (\sec \theta + \tan \theta)^2 d\theta$

- (A) $\frac{(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$
 (B) $\frac{(\sec \theta + \tan \theta)}{3} [2 + 4 \tan \theta (\sec \theta + \tan \theta)] + C$
 (C) $\frac{(\sec \theta + \tan \theta)}{3} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$
 (D) $\frac{3(\sec \theta + \tan \theta)}{2} [2 + \tan \theta (\sec \theta + \tan \theta)] + C$

MII014

15. $\int \left[\sin \alpha \sin(x - \alpha) + \sin^2 \left(\frac{x}{2} - \alpha \right) \right] dx$ equals

- (A) $\frac{1}{2}(x + \sin x) + C$ (B) $\frac{1}{2}(x^2 - \sin x) + C$
 (C) $\frac{1}{2}(x - \sin x) + C$ (D) $\frac{1}{2}(x - \cos x) + C$

MII015

16. $\int \frac{dx}{(x - \alpha)\sqrt{(x - \alpha)(x - \beta)}}$ equals

- (A) $\frac{2}{\alpha + \beta} \sqrt{\frac{x + \beta}{x + \alpha}} + C$ (B) $\frac{-2}{\alpha - \beta} \sqrt{\frac{x - \beta}{x - \alpha}} + C$
 (C) $\frac{-2}{\alpha - \beta} \sqrt{\frac{x + \beta}{x + \alpha}} + C$ (D) $\frac{2}{\alpha + \beta} \sqrt{\frac{x - \beta}{x + \alpha}} + C$

MII016

17. $\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx$ equals

- (A) $(a+x) \arctan \sqrt{\frac{x}{a}} - \sqrt{ax} + C$ (B) $(a+x) \arctan \sqrt{\frac{a}{x^2}} - \sqrt{ax} + C$
 (C) $(a+x^2) \arctan \sqrt{\frac{x^3}{a}} + \sqrt{ax} + C$ (D) $(a+x) \arctan \sqrt{x} + \sqrt{\frac{a}{x}} + C$

MII017

18. $\int \frac{4x^5 - 7x^4 + 8x^3 - 2x^2 + 4x - 7}{x^2(x^2 + 1)^2} dx$ equals

- (A) $4 \ln x - \frac{7}{x} - 6 \tan^{-1}(x) - \frac{6x}{1-x^2} + C$ (B) $2 \ln x + \frac{1}{x} + 6 \tan^{-1}(x^2) + \frac{6x}{1-x^2} + C$
 (C) $\ln x - \frac{1}{x} + \tan^{-1}(x) + \frac{x}{1-x^2} + C$ (D) $4 \ln x + \frac{7}{x} + 6 \tan^{-1}(x) + \frac{6x}{1+x^2} + C$

MII018

19. $\int \frac{3x^2 + 1}{(x^2 - 1)^3} dx$ equals (where K is constant of integration)

(A) $\frac{x}{(x^2 - 1)^2} + K$

(B) $K - \frac{x}{(x^2 + 1)^2}$

(C) $K - \frac{x}{(x^2 - 1)^2}$

(D) $K - \frac{x^2}{(x + 1)^2}$

MII019

20. $\int \frac{x^2 + 3}{x^6(x^2 + 1)} dx$ equals

(A) $C - \frac{2}{x} + \frac{2}{3x^3} - \frac{3}{5x^5} - 2\tan^{-1} x$

(B) $C - \frac{2}{x^2} + \frac{2}{3x^3} - \frac{3}{5x^5} - 2\tan^{-1} x$

(C) $C - \frac{2}{x} + \frac{2}{3x^2} - \frac{3}{5x^4} - 2\tan^{-1} x$

(D) $C - \frac{2}{x^2} + \frac{2}{3x^4} + \frac{3}{5x^6} + 2\tan^{-1} x$

MII020

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. $\int \frac{5x^4 + 4x^5}{(x^5 + x + 1)^2} dx$ equals

- (A) $C - \frac{x+1}{x^5 + x + 1}$ (B) $C + \frac{x}{x^5 + x + 1}$ (C) $C + \frac{x^5}{x^5 + x + 1}$ (D) $C - \frac{x+1}{x^5 - x + 1}$

MII041

2. If $I_n = \int (\sin x)^n dx$ $n \in N$, then $5I_4 - 6I_6$ is equal to -

- (A) $\sin x \cdot (\cos x)^5 + C$ (B) $\cos x \cdot (\sin x)^5 + C$
 (C) $\frac{\sin 2x}{8} [\cos^2 2x + 1 - 2\cos 2x] + C$ (D) $\frac{\sin 2x}{8} [\cos^2 2x + 1 + 2\cos 2x] + C$

MII042

3. Suppose $J = \int \frac{\sin^2 x + \sin x}{1 + \sin x + \cos x} dx$ and $K = \int \frac{\cos^2 x + \cos x}{1 + \sin x + \cos x} dx$. If C is an arbitrary constant of integration then which of the following is/are correct ?

- (A) $J = \frac{1}{2} (x - \sin x + \cos x) + C$ (B) $J = K - (\sin x + \cos x) + C$
 (C) $J = x - K + C$ (D) $K = \frac{1}{2} (x - \sin x + \cos x) + C$

MII043

4. If $f\left(\frac{1-x}{1+x}\right) = x$ and $g(x) = \int f(x) dx$ then

- (A) $g(x)$ is continuous in domain (B) $g(x)$ is discontinuous at two points in its domain
 (C) $\lim_{x \rightarrow \infty} g'(x) = -1$ (D) $\int g(x) dx = -\frac{x^2}{2} + (2x+1) \ln\left(\frac{1+x}{e}\right) + C$

MII044

5. The value of $\int \frac{\ln\left(\frac{x-1}{x+1}\right)}{x^2 - 1} dx$ is equal to

- (A) $\frac{1}{2} \ln^2 \frac{x-1}{x+1} + C$ (B) $\frac{1}{4} \ln^2 \frac{x-1}{x+1} + C$ (C) $\frac{1}{2} \ln^2 \frac{x+1}{x-1} + C$ (D) $\frac{1}{4} \ln^2 \frac{x+1}{x-1} + C$

MII045

6. If $\int \frac{(x-1) dx}{x^2 \sqrt{2x^2 - 2x + 1}} = \frac{\sqrt{f(x)}}{g(x)} + C$, where $f(x)$ is a quadratic expression and $g(x)$ is a monic linear expression.

- (A) $f(x) = 2x^2 - 2x + 1$ (B) $g(x) = x + 1$
 (C) $g(x) = x$ (D) $f(x) = 2x^2 - 2x$

MII046

7. If $\int \frac{4e^x + 6e^{-x}}{9e^x - 4e^{-x}} dx = Ax + B \ln |9e^{2x} - 4| + C$, then

- (A) $A + 18B = 16$ (B) $18B - A = 19$ (C) $A - 18B = 17$ (D) $A + 18B = 32$

MII047

8. $\int \frac{x + x^{2/3} + x^{1/6}}{x(1 + x^{1/3})} dx = px^{2/3} + q \tan^{-1}(x^{1/6}) + c$, then :-

(A) $p < q$

(B) $p = q$

(C) $pq = 9$

(D) $p > q$

MII048

MATRIX MATCH TYPE QUESTION

9.

List-I

List-II

(P) $\int \frac{x^2(x^6 + x^5 - 1)dx}{(2x^6 + 3x^5 + 2)^2}$

(1) $-\frac{1}{3} \frac{1}{(x^3 - x + 1)} + C$

(Q) $\int \frac{(x^5 + x^4 + x^2)dx}{\sqrt{4x^7 + 5x^6 + 10x^4}}$

(2) $\frac{1}{2}(1 + x^{-2} + x^{-5})^{-2} + C$

(R) $\int \frac{(2x^{12} + 5x^9)dx}{(x^5 + x^3 + 1)^3}$

(3) $-\frac{1}{6}(2x^3 + 3x^2 + 2x^{-3})^{-1} + C$

(S) $\int \frac{x^2 - \frac{1}{3}}{(x^3 - x + 1)^2} dx$

(4) $x \left(\frac{x^3}{25} + \frac{x^2}{20} + \frac{1}{10} \right)^{\frac{1}{2}} + C$

(where C is the constant of integration.)

(A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

(B) $P \rightarrow 3; Q \rightarrow 4; R \rightarrow 2; S \rightarrow 1$

(C) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

MII051

MATCHING LIST TYPE QUESTION

10.

Column-I

Column-II

(A) Let $f(x) = \int x^{\sin x} (1 + x \cos x \cdot \ln x + \sin x) dx$ and $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$

(P) rational

then the value of $f(\pi)$ is

(B) Let $g(x) = \int \frac{1 + 2 \cos x}{(\cos x + 2)^2} dx$ and $g(0) = 0$

(Q) irrational

then the value of $g\left(\frac{\pi}{2}\right)$ is

(C) Let $k(x) = \int \frac{(x^2 + 1)dx}{\sqrt[3]{x^3 + 3x + 6}}$ and $k(-1) = \frac{1}{\sqrt[3]{2}}$ then the value

(R) integral

of $k(-2)$ is

(D) If $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx = \ln(f(x)) + g(x) + C$

(S) prime

(where C is the constant of integration and $f(x)$ is positive),
then $f(0) + g(0)$ is

MII052

EXERCISE (S)

1. Let $\int \frac{f'(x)g(x) - g'(x)f(x)}{(f(x) + g(x))\sqrt{f(x)g(x) - g^2(x)}} dx = \sqrt{m} \tan^{-1} \left(\sqrt{\frac{f(x) - g(x)}{ng(x)}} \right) + C$, where $m, n \in \mathbb{N}$ and ' C ' is constant of integration ($g(x) > 0$). Find the value of $(m^2 + n^2)$. MII122
2. A function g defined for all positive real numbers, satisfies $g'(x^2) = x^3$ for all $x > 0$ and $g(1) = 1$. Compute $g(4)$. MII123
3. Suppose $\int \frac{1 - 7 \cos^2 x}{\sin^7 x \cos^2 x} dx = \frac{g(x)}{\sin^7 x} + C$, where C is arbitrary constant of integration. Then find the value of $g'(0) + g''\left(\frac{\pi}{4}\right)$ MII124
4. If $\int \frac{\cos^5 x}{\sin x} dx = \ell \log |\sin x| + m \sin^4 x + n \sin^2 x + C$, (where C is constant of integration), then $\frac{\ell - n}{m}$ is MII125
5. If the value $\int \frac{1 - (\cot x)^{2008}}{\tan x + (\cot x)^{2009}} dx = \frac{1}{k} \ell n |\sin^k x + \cos^k x| + C$, then find k . MII053
6. Let $f(x)$ is a quadratic function such that $f(0) = 1$ and $\int \frac{f(x)dx}{x^2(x+1)^3}$ is a rational function, find the value of $f'(0)$ MII054
7. If the integral $\int \frac{5 \tan x}{\tan x - 2} dx = x + a \ell n |\sin x - 2 \cos x| + k$ then a is equal to : MII055
8. If $\int \frac{x^4 - 2x^2 - 3}{x^4 \sqrt{x(1+x^{-2})^2 + 1}} dx = 2(a + bx + cx^{-1} + dx^{-3})^{\frac{1}{2}} + K$, where K is the integration constant, then the value of $a + b + c + d$ is MII056
9. Let F is anti-derivative of f defined by $f(x) = 5(x^4 - 1)$ where $F(1) = 20$, then value of $F(2)$ is MII057
10. $\int \frac{dx}{16 + x^2} = f(x)$ where $f(4) = \frac{\pi}{16}$ and $(\pi = 3.14)$, then value of $f(2) + f\left(\frac{4}{3}\right)$ is - MII058

EXERCISE - JEE (Main) PYQ

1. For $x^2 \neq n\pi + 1, n \in \mathbb{N}$ (the set of natural numbers), the integral

$$\int x \sqrt{\frac{2\sin(x^2-1) - \sin 2(x^2-1)}{2\sin(x^2-1) + \sin 2(x^2-1)}} dx \text{ is equal to :}$$

(where c is a constant of integration)

[JEE (Main) 2019]

- (1) $\log_e \left| \sec \left(\frac{x^2-1}{2} \right) \right| + c$ (2) $\log_e \left| \frac{1}{2} \sec^2(x^2-1) \right| + c$
 (3) $\frac{1}{2} \log_e \left| \sec^2 \left(\frac{x^2-1}{2} \right) \right| + c$ (4) $\frac{1}{2} \log_e |\sec(x^2-1)| + c$

MII032

2. If $\int \frac{\sqrt{1-x^2}}{x^4} dx = A(x)(\sqrt{1-x^2})^m + C$, for a suitable chosen integer m and a function $A(x)$, where C

is a constant of integration then $(A(x))^m$ equals :

[JEE (Main) 2019]

- (1) $\frac{-1}{3x^3}$ (2) $\frac{-1}{27x^9}$ (3) $\frac{1}{9x^4}$ (4) $\frac{1}{27x^6}$

MII033

3. The integral $\int \cos(\log_e x) dx$ is equal to : (where C is a constant of integration) [JEE (Main) 2019]

- (1) $\frac{x}{2} [\sin(\log_e x) - \cos(\log_e x)] + C$ (2) $\frac{x}{2} [\cos(\log_e x) + \sin(\log_e x)] + C$
 (3) $x [\cos(\log_e x) + \sin(\log_e x)] + C$ (4) $x [\cos(\log_e x) - \sin(\log_e x)] + C$

MII034

4. If $\int \frac{\cos x dx}{\sin^3 x (1 + \sin^6 x)^{2/3}} = f(x)(1 + \sin^6 x)^{1/3} + c$

where c is a constant of integration, then $\lambda f\left(\frac{\pi}{3}\right)$ is equal to

[JEE (Main) 2020]

- (1) -2 (2) $-\frac{9}{8}$ (3) 2 (4) $\frac{9}{8}$

MII035

5. If $\int \frac{d\theta}{\cos^2 \theta (\tan 2\theta + \sec 2\theta)} = \lambda \tan \theta + 2 \log_e |f(\theta)| + C$ where C is a constant of integration, then

the ordered pair $(\lambda, f(\theta))$ is equal to :

[JEE (Main) 2020]

- (1) $(-1, 1 + \tan \theta)$ (2) $(-1, 1 - \tan \theta)$ (3) $(1, 1 - \tan \theta)$ (4) $(1, 1 + \tan \theta)$

MII036

Indefinite Integration

6. The integral $\int \frac{dx}{(x+4)^{\frac{8}{7}}(x-3)^{\frac{6}{7}}}$ is equal to :

(where C is a constant of integration)

[JEE (Main) 2020]

(1) $\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + C$ (2) $-\left(\frac{x-3}{x+4}\right)^{\frac{1}{7}} + C$ (3) $\frac{1}{2}\left(\frac{x-3}{x+4}\right)^{\frac{3}{7}} + C$ (4) $-\frac{1}{13}\left(\frac{x-3}{x+4}\right)^{\frac{13}{7}} + C$

MII037

7. If $\int \frac{\cos x - \sin x}{\sqrt{8 - \sin 2x}} dx = a \sin^{-1} \left(\frac{\sin x + \cos x}{b} \right) + c$, where c is a constant of integration, then the ordered pair (a, b) is equal to :

[JEE (Main) 2021]

(1) $(-1, 3)$ (2) $(3, 1)$ (3) $(1, 3)$ (4) $(1, -3)$

MII038

8. $\int \frac{(x^2 + 1)e^x}{(x+1)^2} dx = f(x)e^x + C$, Where C is a constant, then $\frac{d^3 f}{dx^3}$ at $x = 1$ is equal to :

[JEE (Main) 2022]

(1) $-\frac{3}{4}$ (2) $\frac{3}{4}$ (3) $-\frac{3}{2}$ (4) $\frac{3}{2}$

MII039

9. For $I(x) = \int \frac{\sec^2 x - 2022}{\sin^{2022} x} dx$ if $I\left(\frac{\pi}{4}\right) = 2^{1011}$, then

[JEE (Main) 2022]

(1) $3^{1010} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$ (2) $3^{1010} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$
(3) $3^{1011} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$ (4) $3^{1011} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

MII040

10. For, $\alpha, \beta, \gamma, \delta \in \mathbb{N}$, if $\int \left(\left(\frac{x}{e} \right)^{2x} + \left(\frac{e}{x} \right)^{2x} \right) \log_e x dx = \frac{1}{\alpha} \left(\frac{x}{e} \right)^{\beta x} - \frac{1}{\gamma} \left(\frac{e}{x} \right)^{\delta x} + C$,

Where $e = \sum_{n=0}^{\infty} \frac{1}{n!}$ and C is constant of integration, then $\alpha + 2\beta + 3\gamma - 4\delta$ is equal to:

[JEE (Main) 2023]

(1) 1 (2) -4 (3) -8 (4) 4

MII064

11. Let $f(x) = \int \frac{2x}{(x^2 + 1)(x^2 + 3)} dx$.

If $f(3) = \frac{1}{2}(\log_e 5 - \log_e 6)$, then $f(4)$ is equal to

[JEE (Main) 2023]

(1) $\frac{1}{2}(\log_e 17 - \log_e 19)$ (2) $\log_e 17 - \log_e 18$
(3) $\frac{1}{2}(\log_e 19 - \log_e 17)$ (4) $\log_e 19 - \log_e 20$

MII065

12. Let $I(x) = \int \sqrt{\frac{x+7}{x}} dx$ and $I(9) = 12 + 7 \log_e 7$. If $I(1) = \alpha + 7 \log_e (1+2\sqrt{2})$, then α^4 is equal to _____.

[JEE (Main) 2023]

MII066

13. If $\int \sqrt{\sec 2x - 1} dx = \alpha \log_e \left| \cos 2x + \beta + \sqrt{\cos 2x \left(1 + \cos \frac{1}{\beta} x \right)} \right| + \text{constant}$, then $\beta - \alpha$ is equal to _____.

[JEE (Main) 2023]

MII067

14. Let $I(x) = \int \frac{x^2 (x \sec^2 x + \tan x)}{(x \tan x + 1)^2} dx$. If $I(0) = 0$ the $I\left(\frac{\pi}{4}\right)$ is equal to

[JEE (Main) 2023]

(1) $\log_e \frac{(\pi+4)^2}{16} - \frac{\pi^2}{4(\pi+4)}$

(2) $\log_e \frac{(\pi+4)^2}{16} + \frac{\pi^2}{4(\pi+4)}$

(3) $\log_e \frac{(\pi+4)^2}{32} - \frac{\pi^2}{4(\pi+4)}$

(4) $\log_e \frac{(\pi+4)^2}{32} + \frac{\pi^2}{4(\pi+4)}$

MII068

15. If $\int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} dx = \frac{1}{12} \tan^{-1}(3 \tan x) + \text{constant}$, then the maximum value of $a \sin x + b \cos x$, is :

[JEE (Main) 2024]

(1) $\sqrt{40}$

(2) $\sqrt{39}$

(3) $\sqrt{42}$

(4) $\sqrt{41}$

MII126

16. If $\int \frac{1}{\sqrt[5]{(x-1)^4 (x+3)^6}} dx = A \left(\frac{\alpha x - 1}{\beta x + 3} \right)^B + C$, where C is the constant of integration, then the value of $\alpha + \beta + 20AB$ is _____.

[JEE (Main) 2024]

MII127

17. Let $I(x) = \int \frac{6}{\sin^2 x (1 - \cot x)^2} dx$. If $I(0) = 3$, then $I\left(\frac{\pi}{12}\right)$ is equal to :

[JEE (Main) 2024]

(1) $\sqrt{3}$

(2) $3\sqrt{3}$

(3) $6\sqrt{3}$

(4) $2\sqrt{3}$

MII128

18. For $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, if $y(x) = \int \frac{\operatorname{cosec} x + \sin x}{\operatorname{cosec} x \sec x + \tan x \sin^2 x} dx$ and $\lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} y(x) = 0$ then $y\left(\frac{\pi}{4}\right)$ is equal to

[JEE (Main) 2024]

(1) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(2) $\frac{1}{2} \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(3) $-\frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(4) $\frac{1}{\sqrt{2}} \tan^{-1}\left(-\frac{1}{2}\right)$

MII129

EXERCISE - JEE (Advanced) PYQ

1. $\int \frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}} dx$ is equal to - [IIT JEE 2006]

(A) $\frac{\sqrt{2x^4-2x^2+1}}{x^2} + c$ (B) $\frac{\sqrt{2x^4-2x^2+1}}{x^3} + c$
 (C) $\frac{\sqrt{2x^4-2x^2+1}}{x} + c$ (D) $\frac{\sqrt{2x^4-2x^2+1}}{2x^2} + c$

MII059

2. Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int x^{n-2} g(x) dx$ equals.

(A) $\frac{1}{n(n-1)}(1+nx^n)^{1-\frac{1}{n}} + K$ (B) $\frac{1}{n-1}(1+nx^n)^{1-\frac{1}{n}} + K$ [IIT JEE 2007]
 (C) $\frac{1}{n(n+1)}(1+nx^n)^{1+\frac{1}{n}} + K$ (D) $\frac{1}{n+1}(1+nx^n)^{1+\frac{1}{n}} + K$

MII060

3. Let $F(x)$ be an indefinite integral of $\sin^2 x$. [IIT JEE 2007]

Statement-1 : The function $F(x)$ satisfies $F(x + \pi) = F(x)$ for all real x .

because

Statement-2 : $\sin^2(x + \pi) = \sin^2 x$ for all real x .

- (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (C) Statement-1 is True, Statement-2 is False.
 (D) Statement-1 is False, Statement-2 is True.

MII061

4. Let $I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx$, $J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$. Then, for an arbitrary constant c , the value of $J - I$ equals [IIT JEE 2008]

(A) $\frac{1}{2} \ln \left(\frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^{2x} + 1} \right) + c$ (B) $\frac{1}{2} \ln \left(\frac{e^{2x} + e^x + 1}{e^{2x} - e^x + 1} \right) + c$
 (C) $\frac{1}{2} \ln \left(\frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right) + c$ (D) $\frac{1}{2} \ln \left(\frac{e^{4x} + e^{2x} + 1}{e^{4x} - e^{2x} + 1} \right) + c$

MII062

5. The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals (for some arbitrary constant K)

[JEE (Advanced) 2012]

- (A) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$
- (B) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$
- (C) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$
- (D) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

MII063

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	B	B	D	C	B	A	C	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	B	A	C	C	B	A	D	C	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	
Ans.	A,C	B,C	B,C	A,C	B,D	A,C	A,B	A,C	B	
Que.	10									
Ans.	$A \rightarrow Q, B \rightarrow P, C \rightarrow P, R, S, D \rightarrow P, R$									

EXERCISE (S)

1. 8 2. 13.40 3. 5 4. 8 5. 2010
 6. 3 7. 2 8. 5 9. 46
 10. 0.19 or 0.20

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1 or 3	2	2	1	1	1	3	2	1	4
Que.	11	12	13	14	15	16	17	18		
Ans.	1	64	1	3	1	7	2	4		

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5					
Ans.	D	A	D	C	C					

Important Notes

[illegible]